Sang-Hyeon Park

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RESEARCH INTERESTS

KEYWORDS: Computational materials science / Crystal growth / Interfaces & defects / Ion & polaron transport

My interests focus on (1) extending methodologies to simulate complex phenomena in realistic microstructures across relevant time and temperature scales, and (2) understanding structure-chemistry-property relationships governing charge-transport properties.

EDUCATION

Hanyang University (HYU) *Seoul, Korea*
B.S. in Energy Engineering, Summa Cum Laude (Ranked 1st) Mar. 2020 – Feb. 2024
- Overall GPA: 4.36/4.5 (3.92/4.0) | Major GPA: 4.38/4.5 (Credits taken: 162/97)

PUBLICATION

† equal contribution, * corresponding author *5 Peer-reviewed publications*

[J1] **SH Park**, JH Lim, TY Ahn, HK Yoon, J Lee* and H Roh*, "First-Principles Insights into the Role of 3d Transition-Metal Substituents in Intercalating Li_xFeF_3 Cathode Materials for Lithium-Ion Batteries", *ACS Applied Energy Materials*, 8 (19), 14712-14724, (2025). [Link]

[J2] **SH Park**, D Kang, HK Yoon, Y Choi, and J Lee*, "Data-Driven Design Strategies for Chemically Stabilizing Cathode–Coating Interfaces Employing Interpretable Machine Learning", *Journal of Materials Chemistry A*, 14 (39), 26011-26025, (2026). [Link]
[Featured on the Back Cover; Themed collections: 2026 Journal of Materials Chemistry A HOT Papers]

[J3] **SH Park***, C Youn, JH Lim, G Kim, HK Yoon, Y Choi, and J Lee*, "Stabilizing Interfaces in Magnesium Solid-State Batteries via High-throughput Coating Design", *In Press, Small Methods,* (2026).

[J4] H Kang, H Roh, J Lee, **SH Park**, J Park, H Jeong, HK Yoon, TY Ahn* and Y Choi*, "Comparative study of Ta and Ga doping effects on $Li_7La_3Zr_2O_{12}$ garnet electrolytes for advanced thermal battery applications", *RSC Advances*, 15 (33), 27345-27355 (2025). [Link]

[J5] J Kim†, H Kang†, TY Ahn, **SH Park**, M Kim, SH Kang, H Roh, S Lee, and HS Kim*, "Impact reliability of multilayer stacked-cell thermal batteries using 3D FEM simulation and gas gun test validation", *Journal of Materials Research and Technology*, 42, 5060, (2026). [Link]

4 Preprints under review

[J6] JH Lim, **SH Park**, C Youn, TY Ahn, H Roh, HK Yoon, and J Lee*, "Approaching Intrinsic High-Temperature Performance in Garnet-Type Solid Electrolyte (LLZTO): Microstructure and Surface-Chemistry Engineering for Advanced Salt-free Thermal Batteries", *Submitted*

[J7] **SH Park†,** H Won†, D Kang, S Han, HK Yoon, and J Lee*, "Descriptor-guided strategies for interfacial stabilization between cathodes and solid electrolytes and discovery of out-of-domain buffer composition", *Submitted*

[J8] HR Yu, IW Lee, TY Ahn, WJ Ahn, **SH Park**, CN Im, HK Yoon, Y Choi, and YS Lee*, "Reaction-Controlled Fluorination of FeS_2 Cathodes via FeF_3 Surface Layer Formation for High-Voltage Thermal Batteries", *Submitted*

[J9] J Kim†, **SH Park†**, H Roh, S Lee, HS Kim, and TY Ahn*, "Effects of Compressive and Structural Recovery Characteristics of Thermal-Battery Unit-Cell Pellets on Impact Reliability", *Submitted*

3 Manuscripts in preparation

[J10] **SH Park†***, H Won†, and J Lee, "Theoretical investigation of competing $NaFeF_3$ polymorphs against conversion phases demonstrated by simple mechanochemical synthesis", *Manuscript in preparation* (Expected submission: Winter 2026)

[J11] **SH Park**, CN Im, HR Yu, HK Yoon, and JS Yeo*, "Finite Element Investigation of Thermal Stability in Low Melting Electrolytes for Thermal Batteries", *Manuscript in preparation* (Expected submission: Spring 2027)

[J12] H Lee, J Yoon, **SH Park**, S Lee, E Lee, TJ Shin, Y Kang*, "Proton-Mediated Supramolecular Frustration Stabilizes Organic-Inorganic Hybrid Liquid Crystals", Manuscript in preparation (Expected submission: Winter 2026)

SELECTED PRESENTATIONS

Selected first-author presentations.

[C1] "Data-driven evaluation of interfacial stability in batteries: From first-principles to military applications"
- Oral presentation at the *Korean Institute of Military Science and Technology* Spring meeting, Jeju, Republic of Korea (2026)

[C2] "First-principles-investigated 3d transition metal substitution into Li_xFeF_3"
- Poster presentation at the *Korean Ceramic Society* Spring meeting, Yeosu, Republic of Korea (2025)

PATENTS

[P1] "Method for Manufacturing a Thermal Battery Using Surface Passivation Coating of Cathode Powder for a Thermal Battery and Thermal Battery Manufactured Thereby", KR Patent, 10-3012498, (2026)

RESEARCH AND WORK EXPERIENCE

Agency for Defense Development (ADD), Daejeon, Korea — Jun. 2024 – Present

Research Officer for National Defense (Mandatory Military Service, PIs: Dr. Jongseo Lee and Dr. Yusong Choi)

(1) Computational materials science for energy storage materials

- Investigated the influence of 3d transition-metal substituents on the FeF_3 lattice as a function of lithiation level [J1, C2]
- Optimized Ta- and Ga-doped cubic LLZO structures to analyze the interfacial reactivity with $Li_{13}Si_4$ anode materials [J4]
- Evaluated interfacial reactivity for 40k cathode-coating pairs and developed interpretable ML models for stability prediction [J2, C1]
- Identified stabilizing Mg-containing coating materials by integrating thermodynamic screening with NEB calculations [J3, C1]
- Compared the stability of $NaFeF_3$ polymorphs against conversion phases and solid-state synthesized pnma-$NaFeF_3$ [J10]
- Investigated F-substituted FeS_2 and identified phase separation into FeS_2 | FeF_2, consistent with experimental observations [J8]

(2) Research in military energy storage systems

- Fabricated FeS_2 cathode, eutectic salt (LiCl-KCl), and solid electrolyte pellets for reserve thermal batteries [P1]
- Built a powder atomic layer deposition system for cathode materials to suppress parasitic reactions with molten salt [P1]
- Measured and fitted Nyquist Plots for Li-ion conducting garnets and proton-conducting liquid crystals [J4, J6, J12]
- Adjusted surface properties of Ta-LLZO solid electrolytes by acid etching to improve thermal battery discharge performance [J6]
- Developed a finite-element-based analysis model to analyze the thermal stability of a device-level thermal battery [J11, C1]
- Measured mechanical properties for thermal battery components for finite-element simulations of projectile gas gun test [J5, J9]

Korea Institute of Science and Technology (KIST), Seoul, Korea — Sep. 2023 – Feb. 2024

(3) *Computational design of sulfide solid electrolytes,* *Undergraduate Researcher* (Advisor: Dr. Seungho Yu)

- Proposed the effect of isovalent dopants on the phase stability of Li_6PS_5Cl argyrodite through DFT calculations
- Conducted AIMD to study diffusion properties of argyrodites with varying dopants and anti-site defect concentrations

HONORS AND AWARDS

- **Research Officer for National Defense (ROND)** — May 2024 – Present
  - Selected as one of 20 research officers nationwide for STEM-based national defense research
- **Distinguished Graduate Award, HYU** — Feb. 2024
- **Academic Excellence Award (Top 1%)**, HYU, College of Engineering — Jun. 2023
- **2nd Prize, Hanyang Capstone Design Fair (Thesis Poster Presentation)** — Aug. 2023
  - Thesis: Optimization of Poly(acrylamide) Hydrogel Electrolytes' Mechanical Properties for the Flexible Thermogalvanic Cell
- **2nd Prize, 2023 MiliTECH Challenge, ADD** — Dec. 2023
  - Project: Semiconductor Quantum Dot-Fiber Connection Study for Wired Quantum Cryptography Communication
- **Grand Prize, National Undergraduate Chemical Engineering Design Competition, KIChE** — Nov. 2023
  - Project: Development of highly active adsorbents physically adsorbing toxic chemical agents
- **Grand Prize, Energy and Environment Exploration Competition, Samsung Engineering** — Nov. 2022
  - Project: A study on the performance deterioration of membranes for pressure retarded osmosis power generation
- **National Science and Technology Scholarship**, Korea Student Aid Foundation (KOSAF) — Mar.2022 – Feb. 2024
- **Readership Scholarship**, *Winner, 2023 HYU Reading Quiz Competition* — Dec. 2023
- **Merit-based Scholarship I**, Academic Excellence, HYU — Aug. 2020
- **Merit-based Scholarship II**, HYU, Student President in Dept. of Energy Engineering — Mar. 2020 – Feb. 2022
- **Merit-based Scholarship III**, Community Education Service, HYU — 2023

RELEVANT COURSEWORK

Materials Science / Materials Physics and Properties / Industrial Physical Chemistry / Energy Engineering Thermodynamics / Electrochemistry / Inorganic Material Chemistry / Transport Phenomena / Energy Conversion & Storage Materials / Artificial Intelligence and Machine Learning / Numerical Analysis / Instrumental Analysis / Polymer Chemistry

SKILLS

Computational: Quantum ESPRESSO, VASP, LAMMPS, Python, COMSOL Multiphysics, and ML

Experimental: XRD, EIS, SEM, mechanochemical synthesis, coin-cell fabrication, atomic layer deposition, XPS, and electrochemical testing & cycling

LEADESHIP, TEACHING, AND SERVICES

- **Selected Participant**, Overseas Study Workshop, Korea Foundation for Advanced Studies (KFAS) — 2026 – Present
- **Selected Member**, Young Engineers Honor Society, National Academy of Engineering of Korea — 2022 – Present
- **Undergraduate Mentor**, Samsung Engineering Global Environmental Youth Program — 2023
- **Mathematics Mentor** for local high school students, B-Cycle Educational Mentoring Program, Seoul — 2023
- **Freshmen Tutor**: *General Chemistry and Laboratory I*, HYU — 2023
- **Undergraduate Reporter**, Korea Energy Agency — 2022
- **Student President**, 11th Student Council, Department of Energy Engineering, HYU — 2021